



AIMS

African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 7: Transformation of r.v.'s

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1 Transformation of a r.v.'s

The univariate case

Theorem

Let $y = g(x)$ with g differentiable for all x , and either $g'(x) > 0$ for all x or $g'(x) < 0$ for all x , then the p.d.f. of $Y = g(X)$ is

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Exercises:

- Let $X \sim \mathcal{N}(\mu, \sigma^2)$. What is the distribution of $Y = e^X$?

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Solution: Y has a **lognormal distribution** with parameters (μ, σ^2) .

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma y}} e^{\{-(\log y - \mu)^2 / 2\sigma^2\}}, \quad y \in]0, +\infty[.$$

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- Let $X \sim \mathcal{N}(0, 1)$. What is the distribution of $Y = X^2$?
n.b.: we are not in the theorem conditions (g is not strictly monotone).

Prove that $F_Y(y) = 2\Phi(\sqrt{y}) - 1$.

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Let $X_1, \dots, X_n$ be a n -dimensional continuous r.v.

Let $Y_1, \dots, Y_n$ be a n -dimensional continuous r.v. defined by:

$Y_i = g_i(X_1, \dots, X_n)$, $i = 1, \dots, n$, where g_i has a continuous first partial derivative at all $(x_1, \dots, x_n)$, for all i and are such that, the **Jacobian**:

$$J(\underline{y} \mid \underline{x}) = \det \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \dots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \vdots \\ \frac{\partial g_n}{\partial x_1} & \dots & \frac{\partial g_n}{\partial x_n} \end{bmatrix} \neq 0.$$

Then the p.d.f. of $\underline{Y}$ is $f_{\underline{Y}}(\underline{y}) = f_{\underline{X}}(g_1^{-1}(\underline{y}), \dots, g_n^{-1}(\underline{y})) |J(\underline{x} \mid \underline{y})|$.

Remark: $|J(\underline{x} \mid \underline{y})| = |J(\underline{y} \mid \underline{x})|^{-1}$ where $\underline{x}$ in the last expression is replaced by $g^{-1}(\underline{y})$.

The multivariate case

Exercise

Let X_1, X_2 be two independent r.v.'s each having an exponential distribution with $1/\lambda$.

Are the r.v.'s $Y_1 = X_1 + X_2$ and $Y_2 = X_1/X_2$ independent?

The multivariate case

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Let X_1, X_2 be two independent r.v.'s each having an exponential distribution with $1/\lambda$.

Are the r.v.'s $Y_1 = X_1 + X_2$ and $Y_2 = X_1/X_2$ independent?

Solution: $g_1(x_1, x_2) = x_1 + x_2$ and $g_2(x_1, x_2) = x_1/x_2$.

$$\begin{aligned} J(y_1, y_2 \mid x_1, x_2) &= \det \begin{bmatrix} \frac{\partial}{\partial x_1} g_1(x_1, x_2) & \frac{\partial}{\partial x_2} g_1(x_1, x_2) \\ \frac{\partial}{\partial x_1} g_2(x_1, x_2) & \frac{\partial}{\partial x_2} g_2(x_1, x_2) \end{bmatrix} \\ &= \det \begin{bmatrix} 1 & 1 \\ 1/x_2 & -x_1/x_2^2 \end{bmatrix} = -\frac{x_1 + x_2}{x_2^2}. \end{aligned}$$

$$y_1 = x_1 + x_2 \text{ and } y_2 = x_1/x_2 \text{ hence } x_1 = \frac{y_1 y_2}{1 + y_2} \text{ and } x_2 = \frac{y_1}{1 + y_2}.$$

$$\text{Then } |J(x_1, x_2 \mid y_1, y_2)| = \frac{y_1}{(1 + y_2)^2}.$$

Combinations of r.v.'s

Replacing in the formula from the theorem, we have:

$$\begin{aligned}
 f_{Y_1, Y_2}(y_1, y_2) &= f_{X_1}\left(\frac{y_1 y_2}{1 + y_2}\right) f_{X_2}\left(\frac{y_1}{1 + y_2}\right) \frac{y_1}{(1 + y_2)^2} \\
 &= \lambda \exp\left\{-\lambda \left[\frac{y_1 y_2}{1 + y_2}\right]\right\} \lambda \exp\left\{-\lambda \left[\frac{y_1}{1 + y_2}\right]\right\} \frac{y_1}{(1 + y_2)^2} \\
 &= \lambda^2 y_1 \exp\{-\lambda y_1\} \frac{1}{(1 + y_2)^2}
 \end{aligned}$$

$$\begin{aligned}
 f_{Y_1}(y_1) &= \lambda^2 y_1 \exp\{-\lambda y_1\} \underbrace{\int_0^{+\infty} \frac{1}{(1 + y_2)^2} dy_2}_{=1} \\
 &= \lambda^2 y_1 \exp\{-\lambda y_1\}, \quad y_1 \in \mathbb{R}^+.
 \end{aligned}$$

We recognize a gamma distribution with parameters $(2, \lambda)$.

And $f_{Y_2}(y_2) = \frac{1}{(1 + y_2)^2}$, $y_2 \in \mathbb{R}^+$.

Thus $f_{Y_1, Y_2}(y_1, y_2) = f_{Y_1}(y_1) \times f_{Y_2}(y_2) \implies Y_1$ and Y_2 are independent.

Combinations of r.v.'s

• **Summation of independent gaussian r.v.'s**

Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ and denote $S_n = \sum_{i=1}^n X_i$.

- $E(S_n) = E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = n\mu.$
- $Var(S_n) = Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i) = n\sigma^2.$

Proposition

The summation of independent normal r.v.'s is a normal r.v.

Proof: Moment generating function.

Combinations of r.v.

- **Chi-square distribution**

- The p.d.f. of a Chi-square r.v. with n d.o.f. is the p.d.f of a gamma distribution with parameters $(n/2, 1/2)$ that is:

$$f_X(x) = \frac{2^{-n/2}}{\Gamma(n/2)} x^{n/2-1} e^{-x/2}.$$

- Notation: $X \sim \chi^2(n)$
- The summation of the square of n independent standard normal r.v. follows the **Chi-square** distribution with n degrees of freedom (d.o.f.).

Combinations of r.v.

● **Student's t-distribution**

- Let $Z \sim \mathcal{N}(0, 1)$ and $Y \sim \chi_n^2$ be two independent r.v.'s, then:

$$T = \frac{Z}{\sqrt{Y/n}}$$

follows a **Student's t-distribution** with n degrees of freedom.

- Notation: $Z \sim \mathcal{T}(m)$.
- The p.d.f. of a Student's t-distribution is:

$$f_T(t) = \frac{\Gamma[(n+1)/2]}{\sqrt{n\pi} \Gamma(n/2)} \cdot \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad t \in \mathbb{R}^+.$$

Combinations of r.v.

● Fisher distribution

- Let $Y_1 \sim \chi_{n_1}^2$ and $Y_2 \sim \chi_{n_2}^2$, be two independent r.v.'s, then

$$Y = \frac{Y_1/n_1}{Y_2/n_2}$$

follows a Fisher distribution with (n_1, n_2) degrees of freedom.

- Notation: $Y \sim \mathcal{F}(n_1, n_2)$.
- The p.d.f. of a Fisher distribution with (n_1, n_2) degrees of freedom is:

$$f_Y(y) = \frac{\Gamma[(n_1 + n_2)/2]}{\Gamma(n_1/2) \Gamma(n_2/2)} \cdot \left(\frac{n_1}{n_2}\right)^{n_1/2} \cdot \frac{y^{(n_1-2)/2}}{(1 + n_1 y/n_2)^{(n_1+n_2)/2}}.$$